

Solving Absolute Value Linear Inequalities

TEACHER KEY | Algebra II | Unit 2 | Day 7 | TEK A2.6.F

Objective: I can solve absolute value linear inequalities using the graphing method and the algebraic (compound inequality) method.

Vocabulary

Compound inequality — two inequalities joined by **and** or **or**.

"Or" statement — true if EITHER part is true - used for **>** or **≥**.

"And" statement — true only if BOTH parts are true - used for **<** or **≤**.

Part 1 — The Graphing Method

Compare $y1 = |expression|$ to $y2 = constant$. Where $y1$ is BELOW $y2$, that is the "less than" solution; where $y1$ is ABOVE $y2$, that is the "greater than" solution.

Part 2 — The Algebraic Method

Example 1 (**<**, uses **AND**). $|2x + 3| < 5$

$$-5 < 2x + 3 < 5 \rightarrow -8 < 2x < 2 \rightarrow -4 < x < 1$$

Example 2 (**≥**, uses **OR**). $|2x - 3| + 4 \geq 9$

Isolate: $|2x - 3| \geq 5$

$$2x - 3 \leq -5 \text{ or } 2x - 3 \geq 5 \rightarrow x \leq -1 \text{ or } x \geq 4$$

Part 3 — You Try

1. $|x + 4| \leq 6$ **$-10 \leq x \leq 2$**

2. $|3x - 1| > 8$ **$x < -7/3$ or $x > 3$**

3. $|x - 5| + 2 \geq 9$ **$x \leq -2$ or $x \geq 12$**

Practice

1. $|2x| < 10$ **$-5 < x < 5$**

2. $|x + 1| \geq 3$ **$x \leq -4$ or $x \geq 2$**

3. $3|x - 2| < 12$ **$-2 < x < 6$**

4. Is $x = 0$ a solution to $|x - 3| > 5$? **no, since $|0 - 3| = 3$, and $3 > 5$ is false**

Exit Ticket

Solve $|3x + 2| \geq 7$, and state whether it is an "and" or "or" compound inequality.

$x \leq -3$ or $x \geq 5/3$ ("or", since the inequality is $\geq$)